Lesson 9.1 Guided Notes

Session 1 · Independent and Dependent Variables

I can name the independent and dependent variables in a relationship. · 6.EE.C.9

Word / Spanish	Meaning — complete the blank	Example
independent variable / variable independiente	The _____: the quantity you choose or that comes first.	hours studied
dependent variable / variable dependiente	The _____: it changes because of the input.	quiz score
depends on / depende de	The output _____ when the input changes.	Score depends on study time.

Key idea

Sentence test: the dependent variable _____ the independent variable.

On a graph, the independent variable goes on the _____ and the dependent variable goes on the _____.

Worked example

A bike rental costs $5 for each hour. Name the variables.

The two quantities are hours rented and _____.

The cost changes when the hours change, so cost _____ hours.

Answer: Independent: _____ · Dependent: _____

You try

1. Name the dependent variable in each pair.
a. minutes jogged, calories burned _____
b. total bill, pizzas ordered _____

_____

2. The longer Rosa plays a video game, the more levels she completes. Write a sentence with “depends on.”

_____

Sum it up: I can tell which variable is independent by ___ because ___.

_____

Session 2 · Tables of Values

I can build a table of values for a relationship between two variables. · 6.EE.C.9

Word / Spanish	Meaning — complete the blank	Example
table of values / tabla de valores	A list of _____ matched with their outputs.	x: 1, 2 y: 9, 18
input / entrada	The number you start with. It goes in the _____ column.	hours worked
output / salida	The number you get _____ you use the rule.	boxes packed

Key idea

Use the _____ rule on every input.

With a fee, _____ first, then add the fee.

Worked example

A worker packs 9 boxes each hour, so b = 9h. Find b for h = 1, 2, and 3.

h = 1: 9 × 1 = _____

h = 2: 9 × 2 = _____

h = 3: 9 × 3 = _____

Answer: Pairs: (1, 9), (2, 18), (3, _____)

You try

1. Use the rule y = 5x + 1 to complete the table.

_____

2. A copier prints 35 pages each minute. How many pages?
a. 2 minutes: _____
b. 4 minutes: _____

_____

Sum it up: To build a table of values, I ___ and then ___.

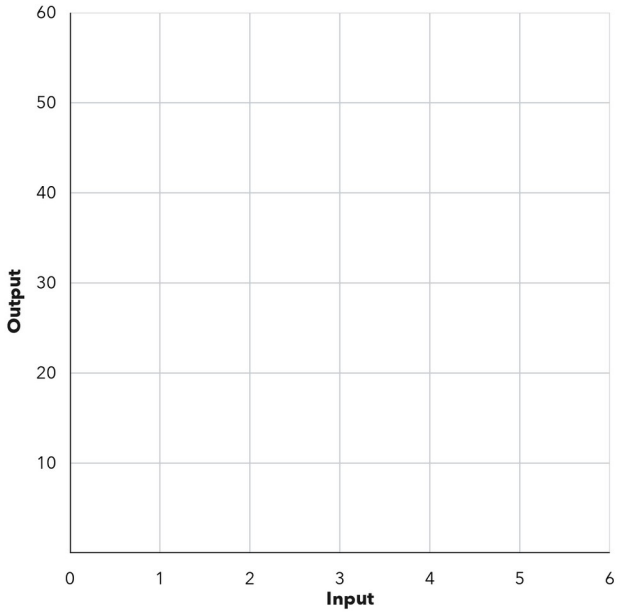
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Session 1 · Model, reason, and explain

Independent and Dependent Variables · 6.EE.C.9

Success: represent the quantities, justify the relationship, and check the result with the correct units.

A shop sells ribbons at $3 per meter with no fixed charge. Define the independent and dependent quantities, explain the dependence, and give the cost for 0,2,5 m.



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Extend your reasoning

Create a contrasting example that exposes the misconception: answers only which axis a variable conventionally uses.. Explain what changes and what stays true.

_____

_____

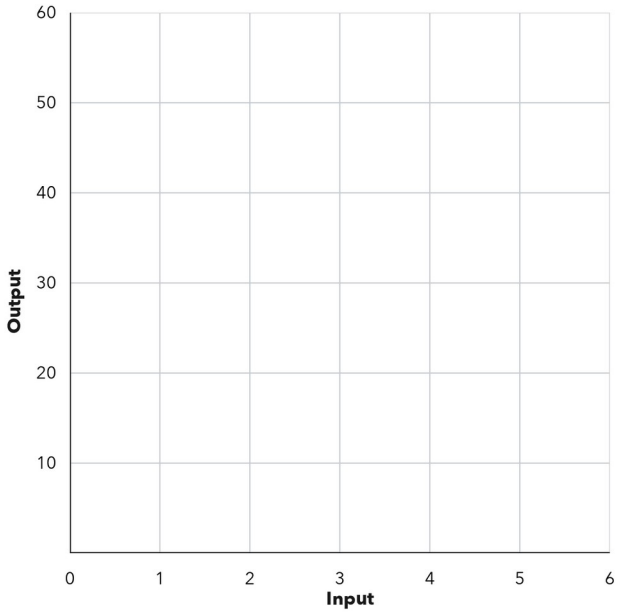
Self-check: Did my model, calculation, explanation, and units agree?

Session 2 · Model, reason, and explain

Tables of Values · 6.EE.C.9

Success: represent the quantities, justify the relationship, and check the result with the correct units.

A delivery costs c=2n+5 dollars for n boxes. Complete n=0,2,5. Explain why adding 2 to each successive output fails for nonconsecutive inputs.



_____

_____

_____

Extend your reasoning

Create a contrasting example that exposes the misconception: adds one rate unit between table rows regardless of input jumps.. Explain what changes and what stays true.

_____

_____

Self-check: Did my model, calculation, explanation, and units agree?